

Semi-Supervised Latent Variable Model for NYC Property Valuation

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STCS 6701: Probabilistic Models and Machine Learning
Professor Blei, Fall 2025

Motivation & Problem

Property valuation underpins **tax administration** to ensure fair revenue and **urban planning** to optimize resource allocation. We target both: taxation requires precise financial estimates, while planning uses value gradients to identify development zones.

Challenges:

- **Heavy-tailed Distribution:** Prices exhibit extreme kurtosis of roughly 9.27, violating standard Gaussian assumptions.
- **Missingness:** About 25% of records lack price data.

Goal: Learn a latent representation to predict sale prices and provide uncertainty estimates without discarding incomplete data.

Methodology

We implement a **Semi-Supervised MIWAE** (SemiSupMIWAE) using a Student- t Mixture Prior.

1. Generative Model

- **Latent Space:** $z_i \sim \sum_{k=1}^K \pi_k t_{\nu}(\mu_k, \Sigma_k)$.
- **Covariates:** $x_i \mid z_i \sim \mathcal{N}(\mu_x(z_i), \text{diag}(\sigma_x^2(z_i)))$.
- **Price Head:** $y_i \mid z_i \sim \mathcal{N}(\mu_y(z_i), \sigma_y^2(z_i))$.

2. Inference

We maximize the importance-weighted lower bound, the ELBO:

$$\mathcal{L} \approx \mathbb{E}_{q_{\phi}} \left[\log \frac{1}{K_{IW}} \sum_{k=1}^{K_{IW}} \frac{p_{\theta}(x_i, y_i \mid z_i^{(k)}) p(z_i^{(k)})}{q_{\phi}(z_i^{(k)} \mid x_i, y_i)} \right]$$

Graphical Model

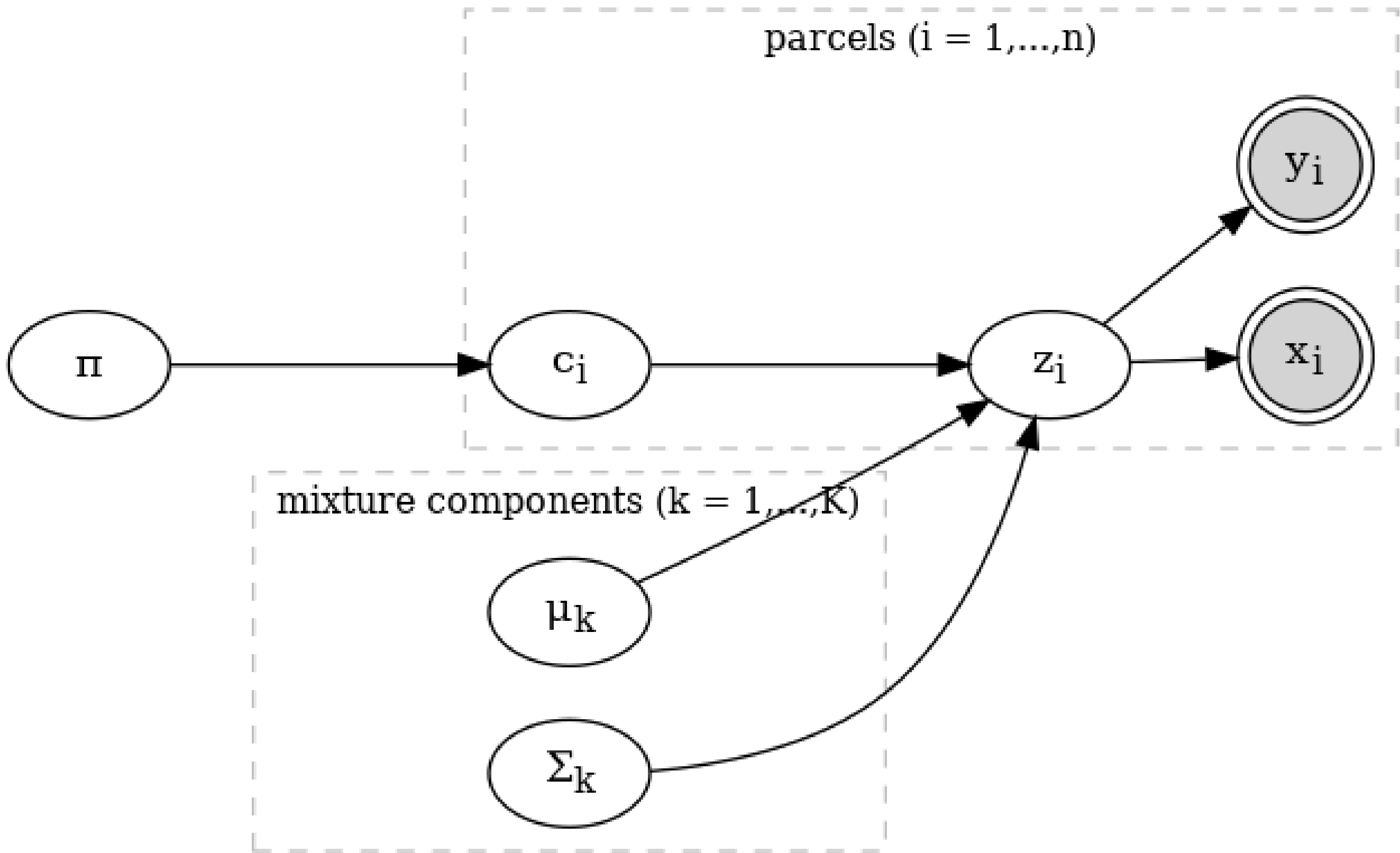


Figure: SemiSupMIWAE Structure where y is semi-observed.

Convergence & Training

Trained on approximately 122,000 NYC parcels. Loss stabilizes around 50 epochs.

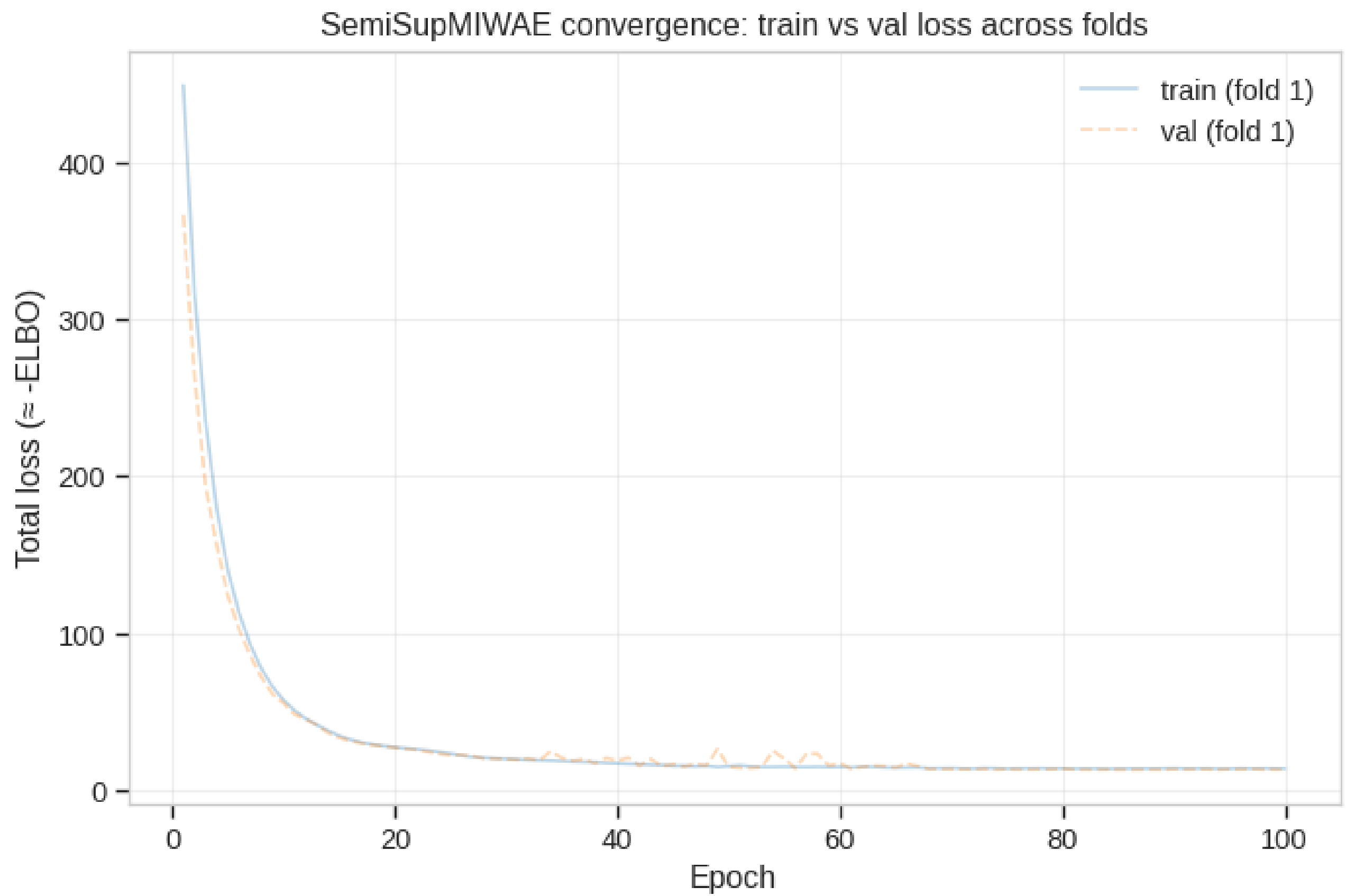


Figure: Training and Validation Loss on a Representative Fold.

Predictive Performance

Evaluated on a 20% held-out set.

Metric	Value
Avg. Log Posterior $p(y \mid x)$	-1.15
RMSE	\$24.6 Million
MAE	\$2.18 Million
Median Absolute Error	\$128,000

Table: Predictive metrics.

Residual Analysis

Residuals show significant heavy tails with Kurtosis approximately 9.27.

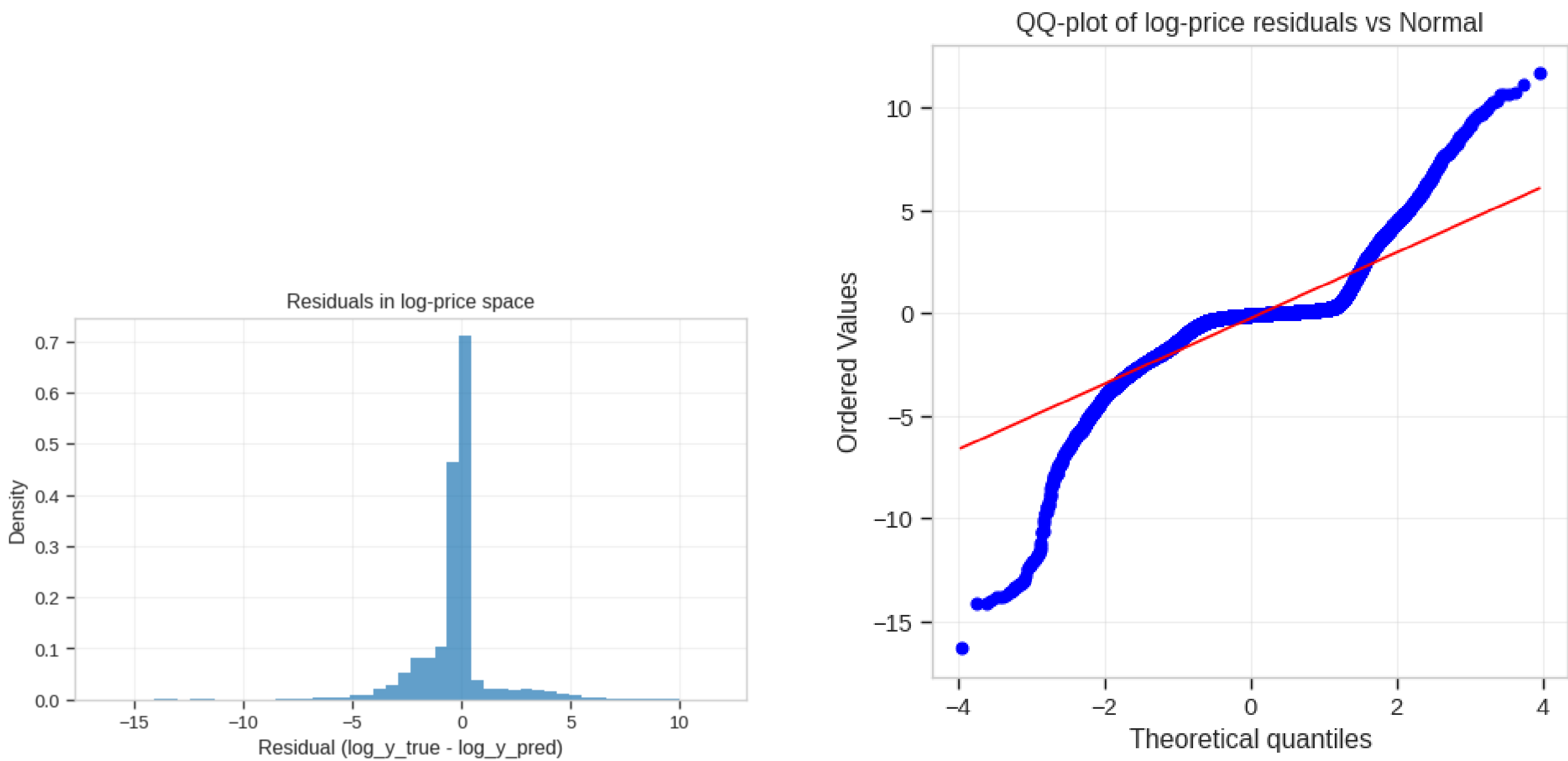


Figure: Log-price residuals vs Normal. Visual comparison highlights extreme kurtosis.

Latent Space Analysis

The 3D latent space, with $K = 3$, captures interpretable structure.

1. Value Continuum

Gradient z_0/z_1 suggests a "Value/Size" factor.

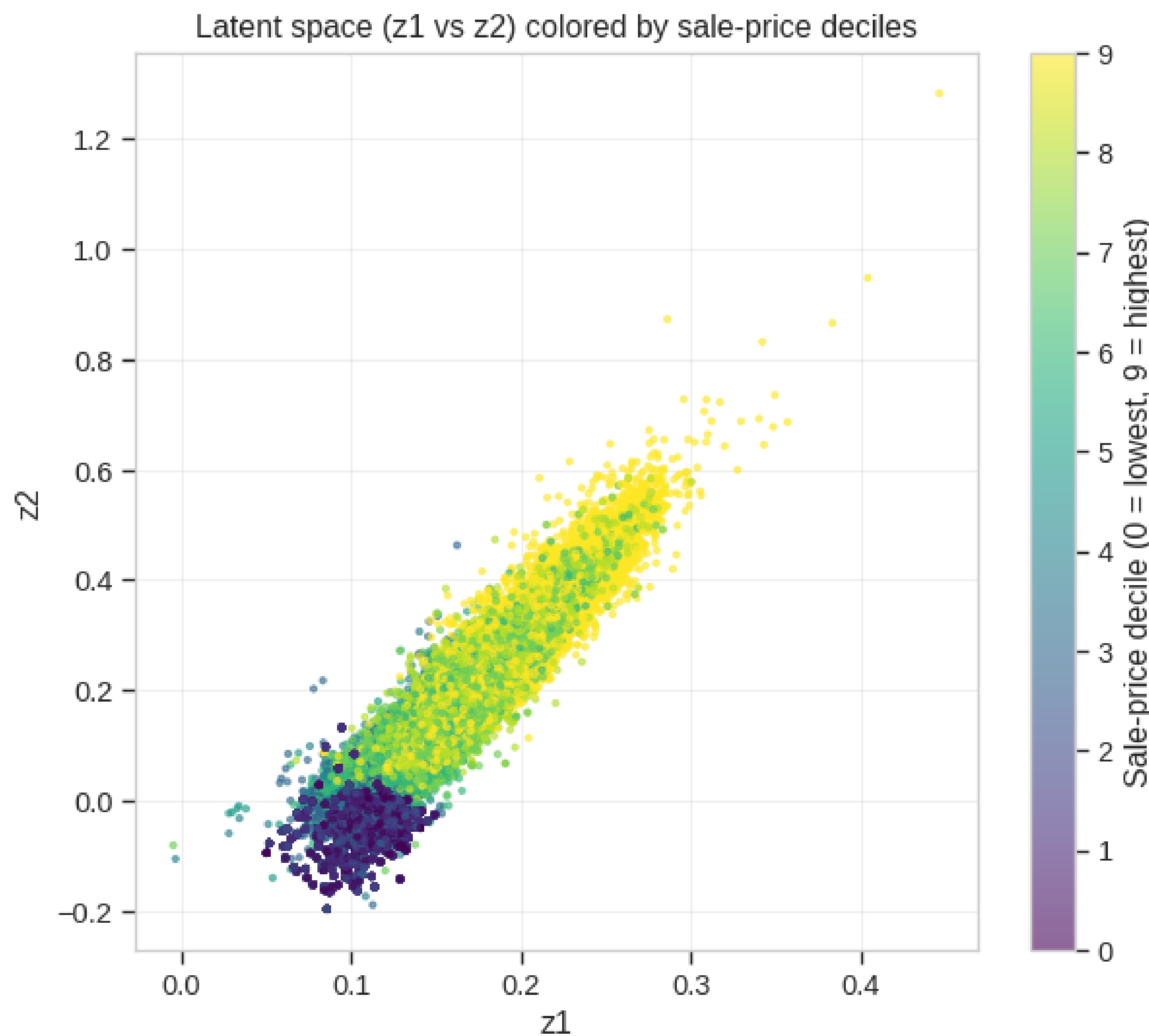


Figure: Latents colored by Price Decile.

2. Structural Clustering

Clusters separate building classes such as Small vs Mixed-Use.

Building class (truncated)

